

ALEKSANDROVSK-SAKHALINSKIY, Russian Federation

WMO: 320610

Lat: 50.900N

Lon: 142.167E

Elev: 31

StdP: 100.95

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.9	(c) -24.2	(d) -28.8	(e) 0.3	(f) -25.4	(g) -27.0	(h) 0.3	(i) -23.8	(j) 14.5	(k) -10.1	(l) 13.0	(m) -9.3	(n) 3.1	(o) 130	(p) 0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.9	(c) 23.0	(d) 18.3	(e) 21.7	(f) 17.6	(g) 20.5	(h) 16.9	(i) 19.5	(j) 21.7	(k) 18.5	(l) 20.7	(m) 17.6	(n) 19.8	(o) 4.5	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.4	20.9	17.6	12.6	20.0	16.7	11.9	19.1	55.9	21.9	52.9	20.8	50.1	19.9	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.2	10.3	8.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-28.9	26.0	2.2	1.3	-30.5	26.9	-31.8	27.7	-33.0	28.4	-34.6	29.4		
			-29.1	20.7	2.2	1.3	-30.7	21.7	-31.9	22.5	-33.2	23.2	-34.8	24.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-15.7	-14.5	-7.9	0.0	6.4	11.5	15.6	16.8	12.8	5.3	-4.1	-12.1		
	DBStd	12.05	5.16	4.75	4.70	3.67	3.40	2.94	2.51	2.49	3.27	3.61	4.84	5.20		
	HDD10.0	3747	796	686	555	300	121	17	0	0	12	153	422	685		
	HDD18.3	6253	1054	919	813	549	370	205	91	62	168	406	672	943		
	CDD10.0	556	0	0	0	1	9	62	173	211	94	6	0	0		
	CDD18.3	21	0	0	0	0	0	0	6	14	1	0	0	0		
	CDH23.3	30	0	0	0	0	1	1	9	19	1	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0		
(14) Wind	WSAvg	3.9	4.1	3.9	4.0	3.9	3.5	3.1	2.9	3.2	3.9	4.6	5.0	4.6		
(15) Precipitation	PrecAvg	622	30	27	24	30	47	37	58	81	95	89	48	55		
	PrecMax	768	70	72	77	89	99	82	153	205	198	224	97	95		
	PrecMin	479	3	4	4	5	10	3	0	4	24	32	7	18		
	PrecStd	80	18	16	16	18	22	21	31	47	34	43	22	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.1	0.2	4.0	14.2	19.1	22.4	24.6	25.2	22.4	16.8	9.3	2.2		
		MCWB	-1.3	-1.5	0.3	7.4	11.1	15.0	18.2	20.0	17.0	11.9	6.0	0.6		
	2%	DB	-2.5	-2.9	2.1	10.2	16.3	20.1	22.5	23.5	20.7	14.5	6.6	-0.2		
		MCWB	-3.4	-4.1	-0.4	4.7	9.7	14.3	17.9	19.0	16.0	10.8	4.5	-1.1		
	5%	DB	-5.0	-4.8	0.5	7.4	14.0	18.2	21.1	22.2	19.4	12.6	4.3	-2.1		
		MCWB	-5.8	-5.9	-1.4	3.5	8.8	13.4	17.3	18.4	15.5	9.3	2.4	-3.0		
	10%	DB	-7.8	-6.9	-0.8	5.3	12.1	16.5	19.8	21.0	18.1	11.0	2.5	-4.2		
		MCWB	-8.5	-7.8	-2.5	2.2	8.0	12.9	16.7	17.8	14.8	8.2	0.8	-5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.2	-1.4	1.2	8.1	12.6	16.5	19.9	21.4	18.2	13.1	6.8	0.7		
		MCDB	-0.3	0.2	2.9	13.0	17.2	20.0	23.1	23.9	20.6	15.4	8.4	1.6		
	2%	WB	-3.3	-4.0	-0.2	5.3	10.8	15.2	18.7	20.1	17.1	11.2	4.5	-1.0		
		MCDB	-2.5	-3.0	1.5	9.1	14.7	18.4	21.2	22.3	19.4	13.8	6.5	-0.3		
	5%	WB	-5.8	-5.8	-1.2	3.7	9.6	14.2	17.9	19.2	16.1	9.8	2.5	-2.9		
		MCDB	-5.1	-4.9	0.4	7.0	13.1	17.3	20.2	21.2	18.5	12.1	4.1	-2.0		
	10%	WB	-8.4	-7.8	-2.5	2.4	8.5	13.2	17.1	18.3	15.1	8.5	1.0	-5.1		
		MCDB	-7.8	-6.8	-0.9	5.0	11.4	15.9	19.3	20.4	17.8	10.7	2.3	-4.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.8	7.5	6.1	6.9	6.7	6.0	5.9	7.1	6.1	5.0	5.7		
		MCDBR	7.1	7.9	7.3	9.5	10.5	9.8	7.9	7.2	8.3	8.2	6.7	6.8		
	5% WB	MCWBR	6.8	7.1	5.8	5.6	5.7	5.3	4.4	4.1	5.2	5.9	5.7	6.3		
		MCWBR	7.0	7.5	6.8	8.6	8.8	8.2	6.4	5.8	7.2	7.5	6.5	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.331	0.405	0.511	0.553	0.507	0.485	0.447	0.395	0.413	0.371	0.310		
	taud		2.010	1.928	1.802	1.720	1.773	1.992	2.123	2.254	2.328	2.080	2.033	2.021		
	Ebn at Noon		679	756	760	723	717	760	772	785	788	661	575	588		
	Edh at Noon		87	124	172	211	209	170	147	123	102	108	87	74		
(44) All-Sky Solar Radiation	RadAvg		1.32	2.46	3.74	4.44	4.84	5.32	4.83	4.08	3.36	2.04	1.15	0.91		
	RadStd		0.11	0.15	0.26	0.43	0.44	0.54	0.56	0.35	0.28	0.18	0.13	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page